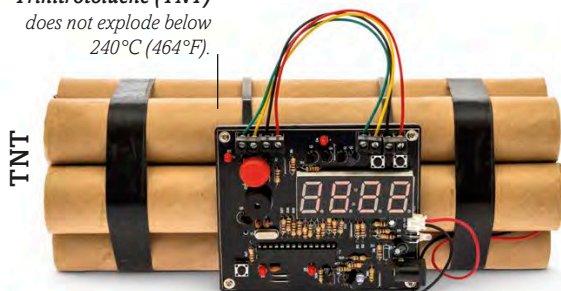


Uses

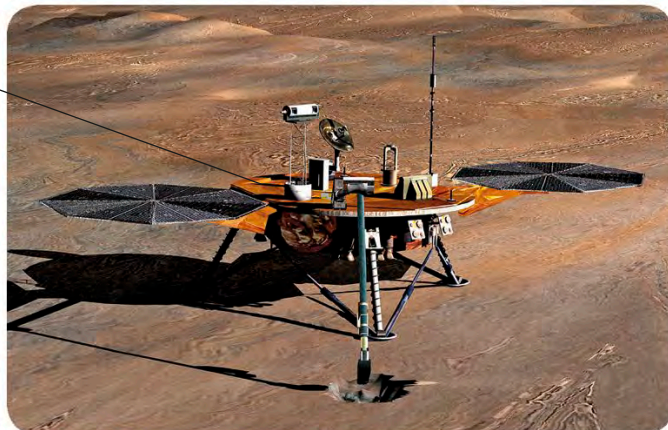
Trinitrotoluene (TNT)
does not explode below
240°C (464°F).



TNT

*This lander used
12 hydrazine-powered
thrusters to land on Mars.*

Phoenix Mars Lander



Nitroglycerine
can be used
to treat heart
conditions.



Nitroglycerine spray



Drag bike

*Motorcycles use
nitromethane to fuel
their powerful engines.*

*These nitrogen-based azo dyes
are often used to colour textiles.*



Textile dye

*Superstrong glues contain small
nitrogen compounds that link together.*



Super glue

*These fertilizers contain compounds called
ammonium nitrates that promote growth in plants.*

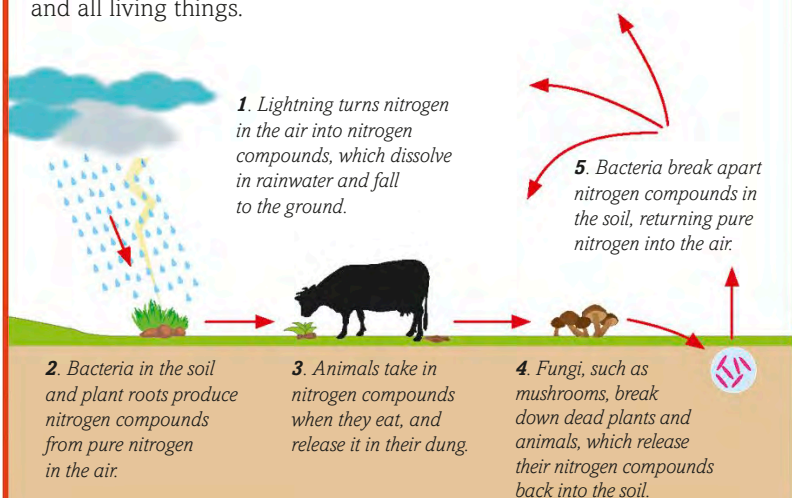


Nitrogen fertilizer



NITROGEN CYCLE

Nitrogen is essential for life. The nitrogen cycle is a process that constantly recycles nitrogen between Earth's atmosphere and all living things.



between nitrogen atoms detach very quickly. Nitrogen fuels, such as nitromethane, are used in **drag bikes**, providing a lot more power than carbon-and-hydrogen only fuels, such as petrol. The compound hydrazine is used in thrusters on spacecraft, such as the **Phoenix Mars Lander**.

Some nitrogen compounds are put in **dyes** and **glues**. An industrial technique called the Haber process turns nitrogen and hydrogen gas into ammonia, a liquid commonly used to make **nitrogen fertilizers**. When mixed with soil, these fertilizers boost plant growth.